International College of Economics and Finance

International Programme in Economics and Finance

Algorithmic Detection of Optimal Stopping Points in Financial Markets Based on the Changepoint Model

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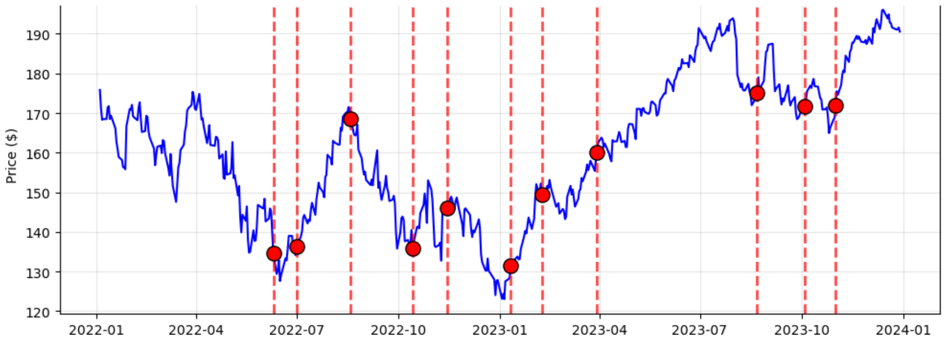
Moscow 2026

Motivation and research aim

- An investor holds a position while the market trend can reverse at a random, unannounced moment.
- Exit too early and profit is lost. Exit too late and the drawdown is taken in full.
- The change point is the moment when the statistical properties of returns shift. Detection has to be sequential: at every date the decision uses only the data available up to that date.
- **Aim:** to develop and test an algorithmic approach to detecting optimal stopping points in financial markets based on the changepoint model, suitable for implementation in Python and for sequential data processing.
- **Object:** financial time series.
- **Subject:** changepoint detection methods applied to exit timing.



AAPL closing price, 2022-2024, with detected change points (Figure 12 of the thesis)

What was done in this work

5 + 2

Five complementary detectors and two control charts, implemented from scratch in Python.

1 rule

A consensus exit rule, the new element of the work: a point is declared only when at least two detectors agree within 15 days.

5 assets

Calibrated on AAPL, then applied without modification to the S&P 500, its futures, TLT and VIX.

Theoretical basis: Zhitlukhin and Ziemba (2016) and Shiryaev (2017). The full code is open: the GitHub link is given in the abstract of the thesis.

Theoretical foundation

- Base article: Zhitlukhin and Ziemba (2016), exit strategies in bubble-like markets using a changepoint model.
- Price model: Gaussian log-returns whose mean and volatility switch at an unknown random moment ϑ.
- The Shiryaev-Roberts statistic accumulates the likelihood of a change; the alarm moment is the exit signal (Shiryaev, 2017; Page, 1954).
- Applied to the crashes of 1929, 1987, 2008 and 2015, the rule exits before the main fall with moderate losses from the peak.
- My step: from a single statistic to a system of five online detectors with a consensus rule, tested across asset classes.

$$X_t = \mu_1 + \sigma_1 \xi_t,\ \ t < \theta, \qquad X_t = \mu_2 + \sigma_2 \xi_t,\ \ t \geq \theta, \qquad \xi_t \sim \mathcal{N}(0,1)$$

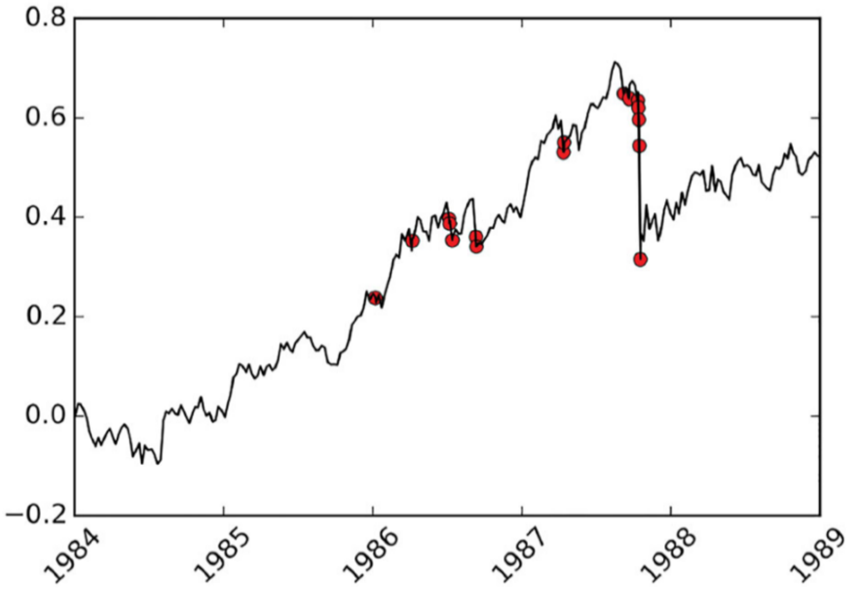


Figure 2. The S&P 500 index in 1984–1988 (logarithmic scale) and the exit points.

Zhitlukhin and Ziemba (2016): exit points before the 1987 crash, S&P 500, log scale

Data: AAPL daily prices, 2022-2024

Sample

- Apple Inc. closing prices, NASDAQ, 1 January 2022 to 1 January 2024.
- 500 daily observations, downloaded with the yfinance Python library.
- Price range over the period: $122.93 to $195.89.
- Windows: 20 days for detection, 60 days for estimating μ and σ. Strictly online regime: only data up to t.
- Log-returns:

$$X_t = \ln(S_t / S_{t-1})$$

Statistical properties of log-returns

- Mean daily return 0.0135%, standard deviation 1.8276%.
- ADF and KPSS agree: the series is stationary.
- Ljung-Box on returns: no linear autocorrelation ($p = 0.54$).
- Jarque-Bera rejects normality ($p = 0.000004$), excess kurtosis 1.84: heavy tails.
- ARCH-LM: volatility clustering is present.

These are the classical stylized facts of returns (Cont, 2001): the Gaussian i.i.d. assumptions hold only in part, and the design of the system takes this into account.

Detection system: five detectors plus control charts

Detector	Reacts to	Threshold	Threshold basis
Score-based log-CUSUM	sustained mean shifts	$T > 5$	empirical: false alarm rate on calm segments
Two-sided CUSUM	drift in either direction	$\lvert C \rvert > 4$	empirical calibration of the detection record
Canonical Shiryaev-Roberts	accumulated likelihood evidence	$\psi > 105.66$	$ARL_0 = 100$ via $h - 1 - \ln h = T$
Cohen effect-size ratio	large window-to-window shifts	$E \geq 0.8$	boundary of a large effect (Cohen, 1988)
Sharpe-like z-score	unfavourable risk-return state	$Z < -0.5$	half-sigma exit sensitivity

Supporting layer: moving-average and EWMA control charts with three-sigma limits. All statistics run online and reset to zero after each alarm, with a 30-day cooldown between detections.

Changepoint model and core detectors

Return model with an unknown change point θ

$$X_t = \mu_1 + \sigma_1 \xi_t, \ \ t < \theta, \qquad X_t = \mu_2 + \sigma_2 \xi_t, \ \ t \geq \theta, \qquad \xi_t \sim \mathcal{N}(0,1)$$

Canonical Shiryaev-Roberts statistic, multi-cyclic optimal (Pollak, 1985; Shiryaev, 2017)

$$\psi_n = \sum_{\theta=1}^{n} \frac{L_n}{L_{\theta-1}}, \qquad \psi_n = \frac{L_n}{L_{n-1}}(1 + \psi_{n-1}), \quad \psi_0 = 0 \qquad \ln \frac{L_n}{L_{n-1}} = \frac{\delta}{\hat{\sigma}_0}(X_n - \hat{\mu}_0) - \frac{\delta^2}{2}, \qquad \delta = 1.0$$

$$h - 1 - \ln h = T, \qquad T = \mathrm{ARL}_0 = 100 \ \Rightarrow \ h \approx 105.66$$

Score-based log-CUSUM, fast worst-case detection (Page, 1954; Lorden, 1971)

$$T_n = \max(0, \ T_{n-1} + \Lambda_n), \qquad \Lambda_n = \frac{X_n - \hat{\mu}_n}{\hat{\sigma}_n}, \qquad T_n > 5$$

Online parameter estimates from a rolling 60-day window

$$\hat{\mu}_t = \frac{1}{60} \sum_{s=t-60}^{t-1} X_s, \qquad \hat{\sigma}_t^2 = \frac{1}{60} \sum_{s=t-60}^{t-1} (X_s - \hat{\mu}_t)^2$$

Both statistics reset to zero after each alarm; a 30-day cooldown prevents signal clustering.

Complementary detectors

Two-sided CUSUM, shifts in either direction (Page, 1954); alarm when |C| > 4

$$C_t^+ = \max(0,\ C_{t-1}^+ + Z_t), \qquad C_t^- = \min(0,\ C_{t-1}^- + Z_t), \qquad Z_t = \frac{X_t - \hat{\mu}_t}{\hat{\sigma}_t}$$

Cohen effect-size ratio, nonparametric magnitude of a shift (Cohen, 1988); alarm when E ≥ 0.8

$$E_t = \frac{|\mu_2 - \mu_1|}{\max(\sigma_1, \sigma_2)}, \qquad [t-2w,\ t-w) \text{ vs } [t-w,\ t), \quad w = 20$$

Sharpe-like z-score, risk-adjusted exit metric; exit signal when Z < -0.5

$$Z_t = \mathrm{Sharpe}_t - 2\,\mathrm{SNR}_t = \frac{\hat{\mu}_t}{\hat{\sigma}_t} - 2\,\frac{|\hat{\mu}_t|}{\hat{\sigma}_t}$$

The five detectors look at one series from different angles: accumulated likelihood, cumulative drift in either direction, window-to-window effect size and risk-adjusted momentum. This methodological diversity is exactly what the consensus rule exploits.

Assumptions are partly violated - why the detectors still work

What the tests show on AAPL log-returns

- Normality is rejected: Jarque-Bera p = 0.000004, excess kurtosis 1.84.
- Volatility clustering: ARCH effects at all conventional lags.
- Stationarity holds (ADF and KPSS), and there is no linear autocorrelation.

Why the methods still work

1. Robustness. CUSUM and SR keep their asymptotic optimality when parameters are estimated rather than known, and under heavier tails (Lai, 1995; Tartakovsky, Nikiforov and Basseville, 2014). The rolling 60-day estimates are consistent under stationarity.

2. The consensus rule. A tail event that fools one detector is rarely confirmed by a methodologically different one within 15 days.

3. A nonparametric anchor. The Cohen effect-size ratio assumes no distribution at all and cross-checks the parametric detectors.

Consensus rule - the new element of this work

An exit point is declared only when at least two different detectors produce a signal within a 15-day window. Such a moment is a consensus point.

Four voting detectors
log-CUSUM 9, CUSUM 15, Cohen 1, z-score 6: about 31 raw signals

→

Group by proximity
signals that fall within 15 trading days of each other

→

Keep agreed clusters
at least 2 methodologically different detectors

→

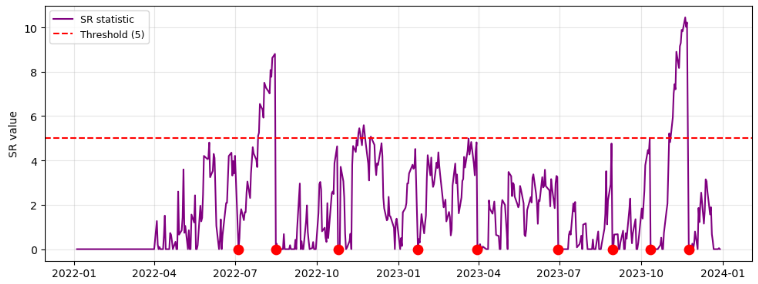
11 consensus
exit points on AAPL

The canonical Shiryaev-Roberts detector stays outside the vote as an independent conservative benchmark.

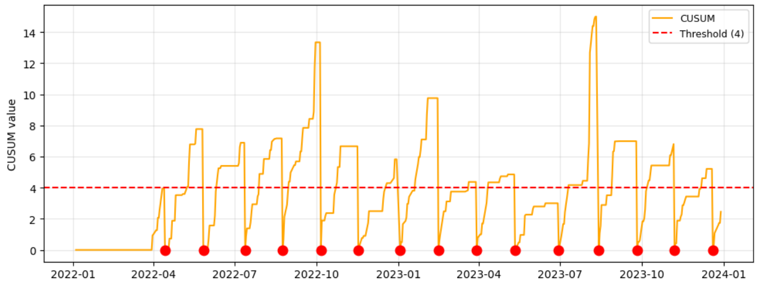
Why two confirmations, not three or four

- The detectors are deliberately selective: on AAPL the Cohen ratio fires once in two years. A three- or four-vote rule would discard signals confirmed by two strong complementary methods.
- Of the 11 consensus points only one (1 July 2022) carries three confirmations. Requiring three votes keeps 1 point of 11 and loses economically justified exits such as 19 August 2022 and 22 August 2023, both close to local peaks.
- Two methodologically different confirmations already do the filtering: about 31 raw crossings become 11 exit points.

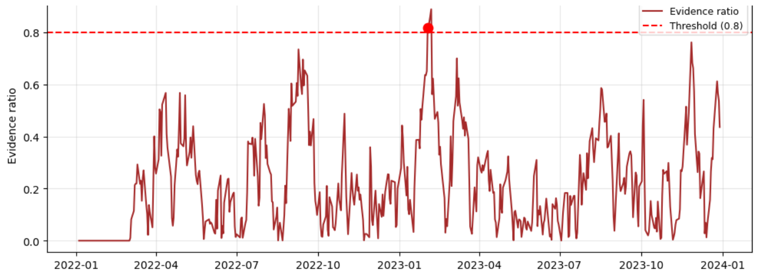
Results on AAPL: individual detectors



Score-based log-CUSUM: 9 crossings (threshold 5)



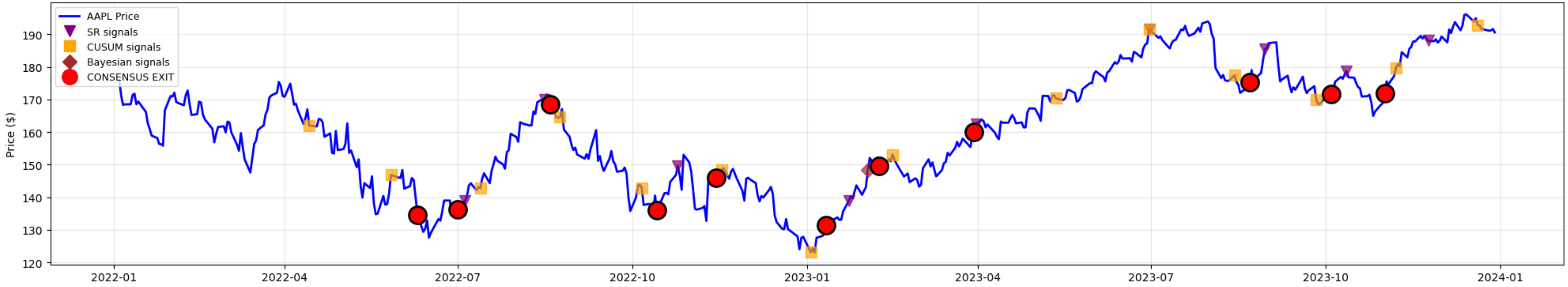
Two-sided CUSUM: 15 crossings (threshold 4)



Cohen effect-size ratio: 1 crossing (threshold 0.8)

- Canonical Shiryaev-Roberts: a single crossing on 10 November 2023 at $184.31, just before the December local peak.
- z-score: 6 stop signals; the deepest on 10 November 2022 at $144.42 with Z = -0.5276.
- Auxiliary t- and F-tests: 72 mean-shift and 49 volatility-shift points; changes in the mean clearly dominate.
- The profiles match the theory: the CUSUM family is sensitive and frequent, SR and Cohen are conservative and precise.

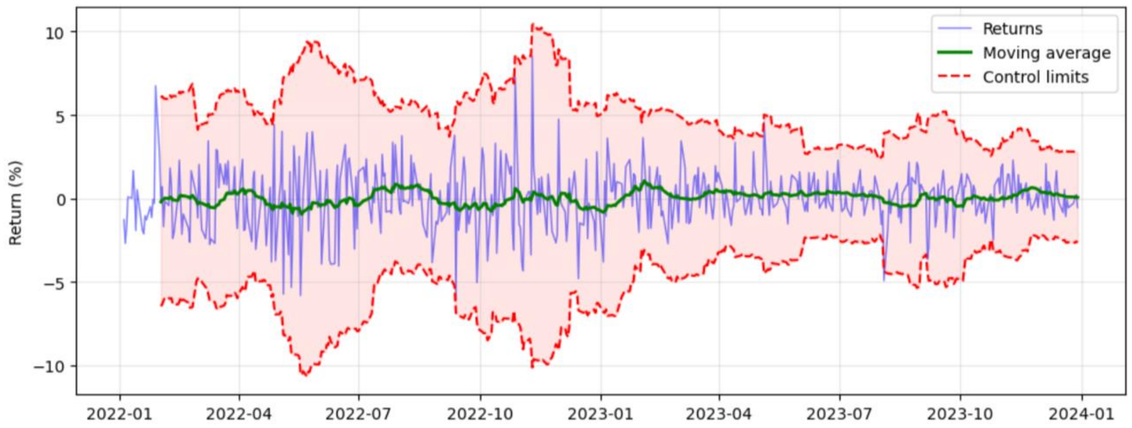
Results on AAPL: 11 consensus exit points



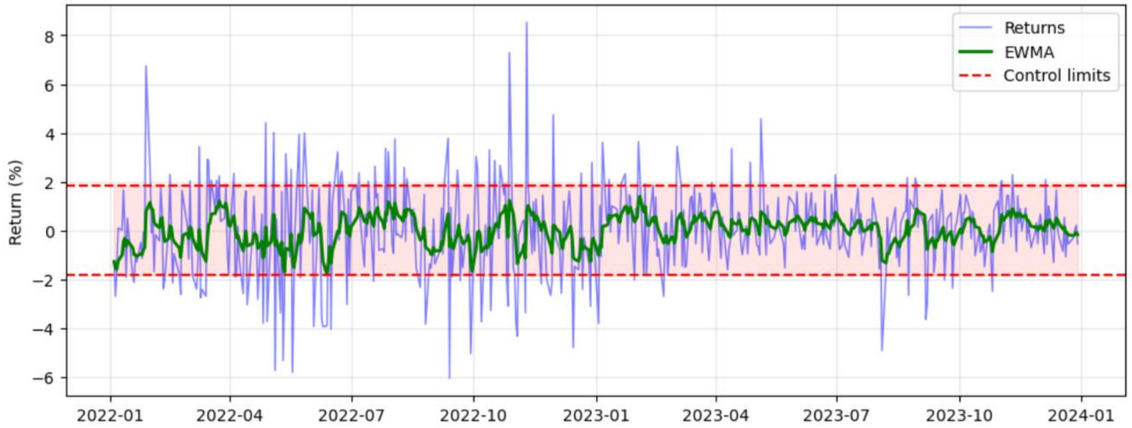
All detector signals and the 11 consensus exit points, AAPL, 2022-2024 (Figure 14 of the thesis)

- The consensus filter compresses about 31 raw crossings into 11 confirmed exit points.
- Ten points are confirmed by two detectors, one (1 July 2022) by three; CUSUM and log-CUSUM appear most often.
- Favourable exits near local peaks: 19 August 2022 at $168.38 and 22 August 2023 at $175.02; 4 October 2023 at $171.49 catches the pullback early.
- Each point lies near an economically meaningful event: Fed decisions, earnings releases, local extrema.

Control charts: a supporting layer



Moving-average chart, 20-day window, ±3σ limits



EWMA chart, α = 0.2, ±3σ limits

$$\mathrm{UCL}_t = \mathrm{MA}_t + 3\,\mathrm{SD}_t, \qquad \mathrm{LCL}_t = \mathrm{MA}_t - 3\,\mathrm{SD}_t$$

$$\mathrm{EWMA}_t = \alpha X_t + (1-\alpha)\,\mathrm{EWMA}_{t-1}, \quad \alpha = 0.2, \qquad \mu \pm 3\sigma\sqrt{\frac{\alpha}{2-\alpha}}$$

Returns outside the bands flag abnormal market states, and the EWMA chart reacts faster to small persistent shifts. The charts visually confirm the detector signals of July and August 2022. They serve as a monitoring layer and do not vote in the consensus.

Robustness: the same model on four more instruments

Instrument	Consensus points	Detector behaviour
AAPL (calibration asset)	11	9 log-CUSUM, 15 CUSUM, 1 Cohen, 6 z-stops
S&P 500 Index	10	9 log-CUSUM, 14 CUSUM, 1 Cohen, 12 z-stops
S&P 500 E-mini Futures	9	mirrors the index within 1-2 trading days
TLT, 20+ year Treasury ETF	11	113 z-stops under negative drift, 0 Cohen crossings
VIX	9	8 z-stops, 0 Cohen: mean reversion, no sharp regime jumps

- The consensus count stays in a narrow 9 to 11 band while daily volatility ranges from 1.22% (futures) to 6.10% (VIX).
- Index and futures trade on different venues, yet the signals coincide: indirect out-of-sample evidence that the model captures the underlying process.
- TLT shows why the consensus matters: the z-score alone fires on roughly one trading day in five, the consensus keeps 11 points.

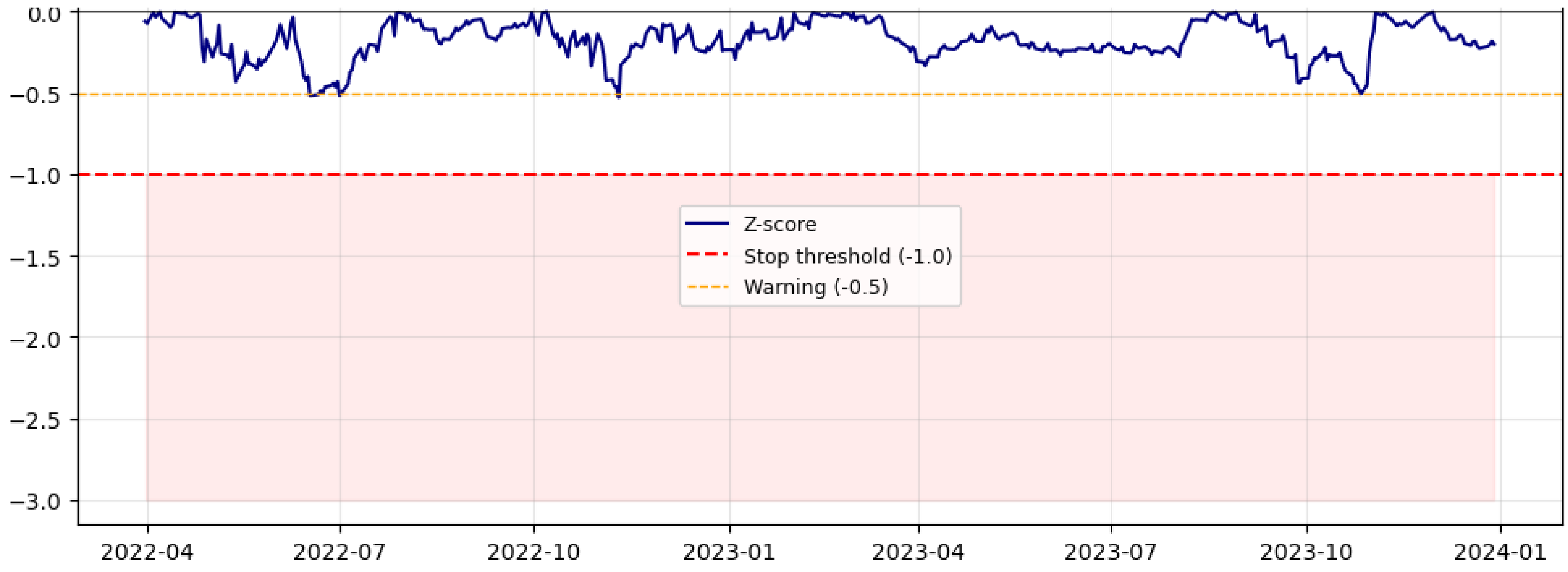
Conclusions

- A complete pipeline was built and tested: the changepoint model, five online detectors, a consensus rule, exit points. Everything is implemented in Python; the code is public and the GitHub link is in the abstract.
- On AAPL the system turns about 31 raw signals into 11 consensus exit points located near economically meaningful events.
- The consensus rule, the new element of the work, keeps false alarms low without losing economically justified exits.
- Across five instruments the number of consensus points stays between 9 and 11: the aggregation logic, not any single detector, drives the result.
- **Limitations:** the i.i.d. Gaussian core assumption, thresholds calibrated on one asset, in-sample evaluation.
- **Next steps:** a GARCH or stochastic-volatility base model, multi-asset recalibration, a formal out-of-sample protocol, transaction costs and trading-rule evaluation.

Thank you for your attention!

Artem Fedosov, ICEF HSE, 2026

Appendix A. The z-score as an exit metric



Z-score for optimal stopping, AAPL, 2022-2024; warning level -0.5 (Figure 8 of the thesis)

- Six stop signals below -0.5 in two years; the deepest on 10 November 2022 at $144.42 (Z = -0.5276, Sharpe ratio -0.1759).
- Other notable stops: 17 June 2022 at $128.97, 1 July 2022 at $136.20, 27 October 2023 at $166.12.
- The signals cluster in mid-2022, the period of the technology drawdown.

Appendix B. Sensitivity of the canonical Shiryaev-Roberts detector

ARL_0 \ δ	δ = 0.5	δ = 1.0	δ = 2.0
50	5	3	2
100	3	1	2
200	0	0	1

Number of detections on AAPL, 2022-2024

- For each target ARL_0 the threshold solves $h - 1 - \ln h = T$ (Lorden-type relation; Shiryaev, 2017).
- The count is non-monotonic in δ: a small δ favours gradual drift, a large δ favours sharp jumps, so an intermediate value can detect fewer events on a series with mixed dynamics.
- The baseline δ = 1.0 with ARL_0 = 100 keeps the detector deliberately conservative: one high-confidence crossing.